

Concern as Fiber Geometry

A Fisher-metric derivation of concern, transport, and holonomy on the stochastic-fibration compiler

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Date: August 3, 2026 **Status:** two theorems + one worked example (finite Boolean world). Companion paper to *The Structural Intelligence Conjecture* ([papers/structural_intelligence/paper.md](#)); depends on Theorem 1 of that paper (existence of the master fibration).

Abstract

The Structural Intelligence Conjecture (SIC) posits a master stochastic fibration (q, κ) : a coarse-graining $q : X \rightarrow Z$ and a compiler kernel $\kappa : Z \rightsquigarrow X$ with $\text{supp } \kappa(\cdot | z) \subseteq q^{-1}(z)$. The companion paper derives (q, κ) from Halmos–Savage minimal sufficiency (Theorem 1) and gives its rate–distortion parameterisation (Theorem 2). This paper adds **concern geometry** to that object: a concern state c is a reweighting of the compiler on the fiber, $\kappa_c(dx | z) \propto e^{\beta U_c(x, z)} \kappa(dx | z)$, and the family $\{\kappa_c\}$ is a **statistical manifold** with a canonical Fisher–Rao metric g_c . We prove two clean theorems from this setup:

- **Theorem CG-1 (Fisher metric on the fiber).** For a smooth exponential-family concern parameterisation, the Fisher information matrix of κ_c on the fiber $q^{-1}(z)$ is the covariance of the sufficient statistic of the concern parameters under $\kappa_c(\cdot | z)$. This makes concern distinctness measurable in bits and gives concern transport its own Riemannian geometry, entirely inside the SIC framework.
- **Theorem CG-2 (Concern holonomy = failure of exactness).** Concern transport around a closed loop of fibers has zero holonomy if and only if the concern-parameterised kernel comes from a *global* potential function on $X \times Z$ (i.e. the concern-1-form is exact). This gives an operational test — the holonomy integral — for whether concern is path-independent (a "flat" concern field) or path-dependent (a non-integrable concern field). Path-dependence is the SIC formalisation of the "meaning shifts under context" intuition.

We close with an exact worked example on the 4-bit Boolean world of Instrument 4: two concern parameterisations of the compiler on the four fibers, an explicit 2×2 Fisher matrix at every parameter value, and a numerical demonstration that a specific non-exact concern field has non-zero holonomy around the natural fiber-loop, while its exact counterpart has zero holonomy.

1. Setup

We take the SIC master object as given: a stochastic fibration (q, κ) with X a standard Borel space, Z a countable coarse space, $q : X \rightarrow Z$ measurable, $\kappa(\cdot | z)$ a probability measure on $q^{-1}(z) \subseteq X$ for each $z \in Z$ (see *The Structural Intelligence Conjecture* §1–§2 for provenance and Theorem 1 for existence).

Concern reweighting. A *concern state* is a parameter $c \in \mathcal{C} \subseteq \mathbb{R}^k$ together with a *concern potential* $U_c : X \times Z \rightarrow \mathbb{R}$ that is jointly measurable and finitely integrable against $\kappa(\cdot | z)$. Concern reweights the compiler:

$$\begin{aligned} \kappa_c(dx | z) &:= \exp(\beta U_c(x, z)) \cdot \kappa(dx | z) / \Xi_c(z), \\ \Xi_c(z) &:= \int_{q^{-1}(z)} \exp(\beta U_c(x, z)) \kappa(dx | z). \end{aligned}$$

$(\beta > 0 \text{ is a fixed inverse-scale})$

The support condition $\text{supp } \kappa_c(\cdot | z) \subseteq q^{-1}(z)$ is preserved: reweighting never crosses fibers. $\Xi_c(z)$ is a *fiber partition function*; requiring $\Xi_c(z) < \infty$ for all z restricts admissible concerns.

Concern manifold. For each z , the map $c \mapsto K_c(\cdot | z)$ gives a parametric family of probability measures on $q^{-1}(z)$ indexed by c . When $c \mapsto U_c(x, z)$ is smooth, this family is a smooth statistical manifold in the sense of Amari–Nagaoka.

2. Theorem CG-1: Fisher information on the fiber

Setup (Theorem CG-1). Fix $z \in Z$. Suppose $U_c(x, z) = \langle c, T(x, z) \rangle$ for some k -dimensional sufficient statistic $T(\cdot, z) : q^{-1}(z) \rightarrow \mathbb{R}^k$. That is, $K_c(\cdot | z)$ is the exponential family with base $K(\cdot | z)$ and sufficient statistic $T(\cdot, z)$:

$$K_c(dx | z) = \exp(\beta \langle c, T(x, z) \rangle - \log \Xi_c(z)) \cdot K(dx | z).$$

Let $E_{\{c, z\}}$ and $\text{Cov}_{\{c, z\}}$ denote expectation and covariance under $K_c(\cdot | z)$.

Theorem CG-1. Under the exponential-family setup, the Fisher information matrix $g_{\{c, z\}}$ on the fiber $q^{-1}(z)$, at parameter c , is

$$g_{\{c, z\}} = \beta^2 \cdot \text{Cov}_{\{c, z\}}[T(\cdot, z)] \in \mathbb{R}^{k \times k}.$$

Consequently the fiber-restricted Riemannian distance between two nearby concerns $c, c + dc$ is $(\beta/1) \cdot \sqrt{(dc^T \cdot \text{Cov}_{\{c, z\}}[T] \cdot dc)}$, and this distance equals (up to leading order) the square-root of twice the Kullback–Leibler divergence

$$KL(K_c(\cdot | z) || K_{\{c+dc\}}(\cdot | z)) = \frac{1}{2} \cdot dc^T \cdot g_{\{c, z\}} \cdot dc + o(\|dc\|^2).$$

Proof. Direct computation. The log-likelihood ratio is

$$\begin{aligned} \log K_c(dx | z) - \log K_{\{c_0\}}(dx | z) \\ = \beta \cdot \langle c - c_0, T(x, z) \rangle - \log \Xi_c(z) + \log \Xi_{\{c_0\}}(z). \end{aligned}$$

The score at c_0 , $\partial_c \log K_c(\cdot | z)|_{\{c=c_0\}}$, is $\beta \cdot (T(\cdot, z) - E_{\{c_0, z\}}[T(\cdot, z)])$ (differentiating $-\log \Xi_c(z)$ gives $-E_c[T]$, standard exponential-family identity). The Fisher matrix is the outer-product expectation of the score:

$$\begin{aligned} g_{\{c_0, z\}} \\ = E_{\{c_0, z\}}[(\beta(T - E_{\{c_0, z\}}[T])) \cdot (\beta(T - E_{\{c_0, z\}}[T]))^T] \\ = \beta^2 \cdot \text{Cov}_{\{c_0, z\}}[T]. \end{aligned}$$

The KL–metric identity $KL(p || p + dp) = \frac{1}{2} dp^T g dp + o(\|dp\|^2)$ is the standard second-order Taylor expansion of KL for smooth families (Amari–Nagaoka, *Methods of Information Geometry*, 2000, Prop. 3.4). $\square$

Consequence (operational). Two concerns are distinguishable in n i.i.d. samples from the fiber iff their KL is at least $\approx 1/n$ (likelihood-ratio test asymptotics); equivalently, iff the geodesic distance between them under $g_{\{c, z\}}$ is at least $\Theta(1/\sqrt{n})$. Concern geometry has a sample-complexity meaning, not just a metaphorical one.

3. Theorem CG-2: Concern holonomy

Cross-fiber concern transport takes the following shape. Fix a base concern $c_0 \in C$. As z varies over Z , the fiber partition function $\Xi_{\{c_0\}}(z)$ varies, and the "canonical" concern-1-form on $Z \times C$ is

$$\alpha(c, z) := \nabla_c \log \Xi_c(z) \, dc = \beta \cdot E_{\{c, z\}}[T(\cdot, z)] \, dc.$$

This 1-form describes how the *mean concern statistic* along a fiber changes as concern varies. Given a closed loop $\gamma : [0, 1] \rightarrow Z \times C$ with $\gamma(0) = \gamma(1)$, the **concern holonomy along γ** is

$$H(\gamma) := \oint_{-\gamma} \alpha = \oint_{-\gamma} \beta \cdot E_{\{c, z\}}[T(\cdot, z)] \, dc.$$

Theorem CG-2 (Concern holonomy vanishes iff concern is exact). $H(\gamma) = 0$ for every closed loop γ if and only if

there exists a smooth function $\Phi : \mathcal{Z} \times \mathcal{C} \rightarrow \mathbb{R}$ such that

$$\alpha = d\Phi, \quad \text{equivalently} \quad \beta \cdot \mathbb{E}_{\{c, z\}}[T(\cdot, z)] = \nabla_c \Phi(z, c).$$

Under smoothness of α (which follows from exponential-family regularity in Theorem CG-1), this is equivalent to

$$\partial_{\{c_i\}}(\beta \mathbb{E}_{\{c, z\}}[T_j]) = \partial_{\{c_j\}}(\beta \mathbb{E}_{\{c, z\}}[T_i]) \quad \text{for all } i, j, z.$$

Proof. Standard de Rham. $H(\gamma) = \oint_\gamma \alpha = 0$ for every closed loop γ iff α is closed ($d\alpha = 0$) on a simply-connected domain, iff α is exact ($\alpha = d\Phi$) by the Poincaré lemma. Closedness in coordinates is the symmetry-of-partial-derivatives condition above (Frobenius / Schwarz). Exponential-family Fisher-metric arguments (Theorem CG-1) show α is smooth in c under $\mathbb{E}_c < \infty$, so the coordinate condition is checkable pointwise. $\square$

Consequence. Non-vanishing holonomy is not an artefact of the parameterisation: it is the *statement* that the concern field cannot be recovered from a single potential Φ on $\mathcal{Z} \times \mathcal{C}$. In SIC terms, the system's concern is *irreducibly path-dependent* — it depends not on the current (z, c) state alone but on how the state was reached. This formalises the intuition (Geometry of Concern; Gauge-Fixed Transport of Concern in the parent repo) that concern deforms under context in ways a single scalar cannot capture.

4. Worked example: the 4-bit Boolean world of Instrument 4

We give a concrete finite calculation exhibiting both theorems.

World. $\mathcal{X} = \{0, 1\}^4$ uniform, $q(x) = (x_0 \oplus x_1, x_2 \oplus x_3) \in \mathcal{Z}$, $|\mathcal{Z}| = 4$. Base compiler $K(\cdot | z)$ is uniform on the 4-element fiber $q^{-1}(z)$.

Concern parameterisation. Take $k = 2$ and

$$\begin{aligned} T(x, z) &= (T_1(x, z), T_2(x, z)) \\ &= (2x_0 - 1, 2x_2 - 1) \in \{-1, +1\}^2. \end{aligned}$$

So concern parameter $c = (c_1, c_2) \in \mathbb{R}^2$ reweights the fiber by $\exp(\beta c_1 (2x_0 - 1) + \beta c_2 (2x_2 - 1))$. Because q fixes only the *parities* of (x_0, x_1) and (x_2, x_3) , each of x_0 and x_2 still ranges over $\{0, 1\}$ on every fiber (the fiber has four elements: two choices for x_0 and, independently, two for x_2 , with x_1 and x_3 then determined by z_1, z_2). So $T_1, T_2 \in \{-1, +1\}$ non-trivially on *every* fiber, and under $K(\cdot | z)$ they are uniform ± 1 and independent — a product of two Rademacher marginals.

(An earlier draft used $T = (x_0 - x_1, x_2 - x_3)$; that statistic is identically 0 on the fibers where $x_0 = x_1$ — half of them — so its Fisher information degenerates there. The choice above is a correction to match the "concern statistic varies on every fiber" requirement of the worked-example setup.)

Fisher matrix (Theorem CG-1 applied). Under $K_c(\cdot | z)$, T_1 and T_2 become independent Rademacher-tilted with means $\mathbb{E}_c[T_1] = \tanh(\beta c_1)$ and $\mathbb{E}_c[T_2] = \tanh(\beta c_2)$, so

$$\begin{aligned} g_{\{c, z\}} &= \beta^2 \cdot \text{diag}(1 - \tanh^2(\beta c_1), 1 - \tanh^2(\beta c_2)) \\ &= \beta^2 \cdot \text{diag}(\text{sech}^2(\beta c_1), \text{sech}^2(\beta c_2)) \end{aligned}$$

for all $z \in \mathcal{Z}$ — the fiber index z really does drop out. Diagonal, depends smoothly on c , with entries in $(0, \beta^2]$. The instrument verifies this exactly at every (c, z) in a pre-registered grid.

A genuinely non-exact concern field. Modify the concern-1-form to

$$\alpha' := \alpha + \varepsilon \cdot (-c_2 \cdot dc_1 + 0 \cdot dc_2).$$

This correction has non-zero curl:

$$\partial_{\{c_2\}}(\alpha'_1) - \partial_{\{c_1\}}(\alpha'_2) = \partial_{\{c_2\}}(-\varepsilon c_2) - 0 = -\varepsilon \neq 0.$$

By Theorem CG-2, α' is not exact, and its holonomy around any closed loop in c -space equals $\varepsilon \cdot (\text{signed area enclosed})$ (Green's theorem; the exact part α integrates to zero by exactness).

(An earlier draft added $\varepsilon \cdot (z_2 \, dc_1 - z_1 \, dc_2)$ at fixed z ; that 1-form is trivially exact on $\mathbb{R}^2$ — its potential is $\Phi(c) = \varepsilon \, z_2 \, c_1 - \varepsilon \, z_1 \, c_2$ — and has zero holonomy on every loop. Detected by the instrument on first run; corrected here to the genuine non-exact form above.)

Numerical holonomy (both loops at fixed $z = (1, 1)$, $\varepsilon = 0.3$).

- Rectangular loop $[0, 1] \times [0, 1]$ — enclosed area 1, so $H(\gamma) = \varepsilon \cdot 1 = 0.3$ (predicted). Instrument computes 0.300000 by trapezoidal quadrature on 500 steps per edge.
- Triangular loop $(0,0) \rightarrow (1,0) \rightarrow (1,1) \rightarrow (0,0)$ — enclosed area $1/2$, so $H(\gamma) = \varepsilon / 2 = 0.15$ (predicted). Instrument computes 0.150000.

Both holonomies non-zero, of *different* magnitudes proportional to the enclosed area — the concrete signature of a path-dependent concern field in the SIC framework, on an exactly solvable finite case.

5. Relation to the SIC framework

Theorem CG-1 is a *direct consequence* of the SIC master object plus exponential-family concern: no new mathematical apparatus beyond Halmos–Savage sufficiency (SIC Theorem 1) and Amari's Fisher-metric identity. It refines the extended-program construct §5.1 of the parent paper ("Concern as fiber geometry") from a *target* to a *theorem*.

Theorem CG-2 is a purely geometric statement about 1-forms on $z \times c$; it does not require probability beyond the smoothness that Theorem CG-1 supplies. Its content is entirely qualitative: it identifies when concern transport is or is not path-dependent, and gives a coordinate condition (symmetry of partial derivatives of $E_c[T]$) that is testable on any concrete concern parameterisation. This refines the extended-program target — the "holonomy measures path-dependence" claim — into an if-and- only-if with a computable check.

Both theorems close targets in the parent paper's §5 extended program: §5.1 "concern as fiber geometry" is now Theorem CG-1 + Theorem CG-2 for the exponential-family + smooth 1-form case. The §5.6 compiler-tomography construct, §5.9 alignment-as-ensemble-governance, and the remaining targets are still open.

6. Limitations

- The exponential-family assumption of Theorem CG-1 is standard but not free: concerns that don't factor as $\langle c, T \rangle$ need a more general score-function argument, still yielding a Fisher matrix but no longer in closed form.
- Theorem CG-2's Poincaré lemma requires simply-connected $z \times c$. For multiply-connected topologies (finite z with a chosen loop structure), the equivalence weakens to "exact $\Rightarrow$ zero holonomy" and holonomy can detect topological non-triviality of z .
- Nothing here upgrades SIC's *cross-task learnability* (SIC-C-b, SIC-C-c in the parent paper) or the *alignment claim* (extended program §5.9). Both remain untouched.
- No machine-checked Lean proof of Theorems CG-1 or CG-2 yet; Amari–Nagaoka's Fisher-metric identity would need mathlib's information geometry (which is minimal). Follow-up.

7. Reproduction

The worked example in §4 is finite and executable. See `experiments/concern_fisher_pair` for an exact enumeration:

```
python3 experiments/concern_fisher_pair/experiment.py
```

Full development is in the parent paper's §5.1 and the notes file `notes/structural_intelligence_conjecture.md`.